

Steven Ngo

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EDUCATION

University of California, San Diego

Bachelor of Science in Data Science, Mathematics Minor, GPA: 3.85/4.0

La Jolla, CA

2024 – 2028

EXPERIENCE

Undergraduate Machine Learning Researcher

Nov 2025 – Present

UCSD Halicioglu Data Science Institute – MixLab

La Jolla, CA

- Evaluated pre- and post-trained LLMs (Llama/Qwen) on long-horizon personalization benchmarks, publishing to **ICML 2026**
- Built a supervised Fine-Tuning (SFT) pipeline to train model params on ToM conversation data, generated by agentic simulators
- Integrated **12** benchmarking and **6** baseline codebases into a unified **LangChain** evaluation framework for vLLM-served models
- Collaborating with **13** domain scientists on autonomous research tools for advanced material science problem-solving

Undergraduate Machine Learning Researcher

Sep 2025 – Feb 2026

UCSD Department of Physics – Rappel Laboratory

La Jolla, CA

- Implemented multi-object tracking algorithms (**ImageJ**, TrackMate) to model **200k+** amoeba merging and splitting events
- Tuned thresholding parameters to detect contours, segment, and vectorize extracted cell tracks across **8000** TIFF frames
- Developed statistical mechanics and particle physics models in Python, using data science libraries (**NumPy**, **pandas**, **Plotly**) to compare cosine, MSD, and velocity distributions of slug migrations towards cAMP chemoattractant waves vs. high-density clusters

Lead Software Engineer

Jul 2025 – Dec 2025

The Bias Lab

San Francisco, CA

- Engineered an end-to-end breaking news analysis platform with **FastAPI** and Next.js, building a scheduled ingestion pipeline fetching **100** NewsAPI articles daily for narrative clustering, securing **\$125k** in pre-seed funding
- Prompt-engineered GPT background jobs that extracted **20+** verified facts and blindspots per story, maintaining **sub-1s** latency via async processing, caching, and FAISS vector database, upserting analyses in S3 and MongoDB for scalable **24/7** on-demand queries
- Increased model precision by **80%** via isolated tests, comparing LLM cost-quality tradeoffs and evaluating K-means vs. HBDSCAN
- Designed **7+** REST API endpoints with Pydantic schema validation, shipping core AI and UX flows via Vercel and AWS CI/CD

Software Engineering Intern

Sep 2025 – Nov 2025

Personalize Send

Arlington, VA

- Built a full React Native-Expo mobile app with Firebase OTP phone auth, OAuth2, Stripe payments, **Gemini-2.5-Flash** GenAI image personalization flow, and lazy-loading/paginated browsing of items to support scalable UX for **500+** early users
- Deployed serverless Cloud Functions, GraphQL APIs, and Stripe Webhooks to handle CRUD operations on account metadata and subscription credit management, whilst linking **200+** products between Firestore database and Shopify users

Algoverse AI Research

Palo Alto, CA

Machine Learning Researcher

Sep 2025 – Nov 2025

- Mentored students on ML research published to **3x AAAI 2026** and **Springer Nature 2026**
- Benchmarked SOTA MLLMs and VLMs on a novel multimodal 2D-to-3D spatial reasoning dataset with Origami crease patterns and fold mappings, evaluating **22** open-source and proprietary models over **3** dimensions (incl. GPT, Claude, Gemini, DeepSeek, Kimi)

Machine Learning Researcher

Jun 2025 – Sep 2025

- First-authored research with PIs (**Meta**, **Berkeley EECS**, **Amazon**) on pivotal token representations in LLM Chain-of-Thought outputs, probing latent features encoding reasoning-shift behavior during inference, accepted to **EMNLP 2025**
- Leveraged NVIDIA **T4/A100** CUDA to train linear and nonlinear classification networks on Qwen3 hidden states, predicting pivotality in mid-late attention heads at **85%** accuracy and diagnosing dataset imbalances via AUROC/F1 and confusion matrices
- Architected data pre-processing and text generation pipelines with Python (**PyTorch**, **pandas**, **sk-learn**, **transformers**), visualizing PCA and t-SNE activation clusters of **29** layers with **Matplotlib/Seaborn** dimensionality reduction

PROJECTS

🏆 League of Legends Statistical Analysis and Predictive Modeling | Python, GitHub Pages

Oct 2025 – Dec 2025

- Cleaned large **200k**-example dataset with **pandas** and engineered features by one-hot encoding and standardization with **sklearn**
- Conducted EDA and hypothesis tests (t-test, permutation) to infer causal relationships between teams on early-game win factors
- Built hyperparameter-tuning and k-fold cross-validation pipelines (**GridSearchCV**) to train **30** ensemble-learning models (Decision Trees/Random Forest) for win prediction, reducing overfitting and optimizing bias-variance tradeoff of Entropy vs. complexity

🏆 SmartRation – UC Berkeley AI Hackathon | TypeScript, Next.js, Express.js, Supabase, Google Vision, Claude

June 2025

- Developed a full-stack computer-vision web app that generates budget-conscious, personalized meal plans from grocery receipts
- Integrated **Google Vision OCR** to parse unformatted receipt data, Kroger API to fetch live market prices, Supabase RLS and email auth, and prompt-engineered Claude to generate shopping lists and dietary recommendations based on user history data

TECHNICAL SKILLS

Languages: Python, SQL, MATLAB, LaTeX, R, Java, C++

Libraries: pandas, scikit-learn, NumPy, PyTorch, Matplotlib, Seaborn, Plotly, OpenCV, HuggingFace, OpenRouter, Sci-Py, FastAPI

Frameworks: Runpod, JUnit, ImageJ, Jupyter/Anaconda, vLLM, PostgreSQL, MongoDB, Tableau, Supabase, Firebase, verl, AGILE

Tools: AWS, Git/GitHub, Docker, Overleaf, Google Cloud, Colab, Vercel, VSCode, IntelliJ, JIRA, Confluence, Notion, MS Excel, Linear